Muhammad Raza

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Education

Kumoh National Institute of Technology — Gumi-si, South Korea
Masters in IT Convergence Engineering; ***GPA: 4.33/4.50*** — *Feb 2023 – Feb 2025*

Karachi Institute of Economics and Technology (PAF-KIET) — Karachi, Pakistan
Bachelors in Computer Science; ***GPA: 3.34/4.00*** — *Aug 2018 – Sept 2022*

Skills

Programming Languages: Python, C++

Machine Learning & Frameworks: PyTorch, TensorFlow, model training and fine-tuning, representation learning

Computer Vision & Perception: 2D/3D object detection, semantic and instance segmentation, object tracking under occlusion, depth estimation, 3D Reconstruction, point cloud processing

Efficient & Trustworthy AI: model compression, quantization, knowledge distillation, LoRA, memory optimization

Multi-Modal Learning: vision–language models (VLMs), vision-action models (VLAs), open-vocabulary perception

Deployment & Systems: edge deployment (Jetson), mobile inference, TensorRT optimization, Docker-based pipelines

Collaboration & Tools: Git, Jira, Confluence, Slack, experiment tracking and deployment workflows

Experience

Neubility Inc. — Seoul, South Korea
R&D AI Robotic Engineer — *Oct 2025 – Present, Full-time*

- **Multi-Camera 2D/3D Tracking Pseudo-Labeling Pipeline:** Designed a geometry-based multi-camera tracking-by-detection system for a 5-camera robot rig, projecting 2D detections to a shared world frame via calibrated depth and per-camera extrinsics to assign globally-consistent track IDs; generated 19K+ pseudo-label annotations across 646 global tracks with zero ID switches on static objects, and extended the detection head with appearance embeddings trained via temporal-consistency and triplet losses.
- **Custom Object Detection Model (Object-Detection with DINOv3):** Built a frozen DINOv3 ViT backbone with a custom FCOS-style detection head extended to a stride-8 pyramid level for small-object recall, trained via a 4-stage mixed COCO+domain curriculum with VariFocal loss and small-target-aware label assignment; achieved mAP@50 0.626 and up to +22pp traffic-light recall via a custom SAHI-based inference augmentation, deployed via split ONNX/TensorRT export to a ROS2 pipeline.
- **End-to-End AI Pipeline for Autonomous Robots:** Designed and deployed efficient end-to-end AI pipelines for the *Neubie* autonomous robot, covering perception, data generation, and deployment-ready inference.
- **2D Auto-Labeling and Pseudo-Label Generation:** Developed a large-scale 2D object detection auto-labeling pipeline to generate high-quality pseudo-labels, significantly reducing manual annotation effort for industrial datasets.
- **3D Object Detection and Deployment:** Fine-tuned state-of-the-art 3D object detection models and adapted them for real-world deployment, enabling reliable 3D bounding box perception on the Neubie robotic platform.
- **3D Dataset Auto-Labeling Tool:** Built an automated 3D labeling framework leveraging open-vocabulary 3D detection, monocular depth estimation, point cloud generation, and 3D scene reconstruction to address large-scale 3D dataset annotation challenges.
- **Core Technologies:** 2D/3D object detection, open-vocabulary perception, depth estimation, point cloud reconstruction, pseudo-labeling, and deployment-oriented model optimization.

3i Inc. — Seoul, South Korea
R&D AI Software Engineer — *Jan 2025 – Oct 2025*

- **PIVO Sports Tracking System:** Developed and deployed end-to-end AI modules for the PIVO sports tracking application, including object detection, single-object tracking, human pose estimation, and real-time inference optimization for consumer devices.

- **Occlusion-Robust Single-Object Tracking:** Designed a lightweight occlusion-handling framework combining fine-tuned object detectors with CSRT-based continuous tracking, enabling stable target localization under partial and full occlusions.
- **Distractor-Aware Memory (DAM):** Implemented a memory-based recovery mechanism for re-detection and identity preservation during occlusion and tracking failure, improving robustness against target switches and visual ambiguity.
- **3D Avatar and Model Optimization:** Contributed to 3D avatar generation pipelines, including 3D mesh simplification and compression for efficient real-time rendering and deployment.
- **Core Technologies:** Object detection, single-object tracking, occlusion handling, human pose estimation, memory-based tracking, model optimization, and real-time deployment.

WENS Lab Gumi-si, South Korea
Research Assistant – Deep Learning and Computer Vision *Feb 2023 – Present, Full-time*

- **Person-Following Autonomous Cart:** Designed and optimized a real-time person-following system for autonomous grocery carts using YOLOv8-based detection and tracking, enabling robust target identification in dynamic environments.
- **UAV-Based QR Code Detection for Smart Factories:** Developed a lightweight QR code detection and recognition pipeline using YOLOv8n integrated with ROS, enabling real-time aerial inspection in smart factory environments.
- **Lightweight Image Classification Architecture:** Proposed and implemented a depthwise separable convolution-based image classification model for efficient multi-class feature extraction across datasets with fewer than 100 object categories.
- **UxV Swarm for Smart Farming:** Developed a UxV swarm-based automated fruit disease instance segmentation framework using YOLOv8, targeting scalable and efficient smart farming applications.
- **Collaborative Indoor 3D Mapping with UxVs:** Contributed to multi-UxV indoor 3D mapping and map-merging systems using VINS-Fusion as the front-end and COVINS-G as the back-end, supporting applications such as coal mining, subway inspection, and smart factories.
- **Industry-Focused Vision Systems:** Developed grid-based person detection algorithms using YOLOv5 with ROS for gimbal-mounted cameras in industrial settings.
- **Additional Applied CV Projects:** Implemented computer vision solutions for industrial use cases including traffic light detection and chemical liquid leakage detection.

1st Author International Journals / Conferences

ASD Submitted June 2026
BMVC Results

- Proposed **Anti-Shortcut Distillation via Temporal Negative Knowledge Transfer**
- Anti-Shortcut Distillation (ASD), a push–pull KD framework that treats the converged teacher Tfinal as a positive semantic anchor and an early-checkpoint teacher Tearly as a temporal negative reference. ASD cou- ples two losses: a temporal contrastive loss (LTC) that places the early-teacher feature as a same-sample negative against in-batch and memory-bank final-teacher features in an InfoNCE objective; and a shortcut suppression loss (LSS) that penalizes student pro- jection onto the top eigenvectors of E[hh], the uncentered second-moment matrix of early-to-final feature displacements.

FSGM-Sum Submitted June 2026
Pattern Recognition Results

- Proposed **FSGM-Sum: Factorized Spatio-Temporal Gating Mixer for Efficient Video Summarization**
- FSGM-Sum, a lightweight Factorized Spatio-Temporal Gating Mixer for single-pass frame importance prediction. Specifically, our method decomposes importance estimation into complementary temporal and channel saliency signals, where temporal dynamics are modeled via multi-dilation depthwise mixing and channel relevance is captured through global pooling with a compact gating network. These signals are fused through an additive factorization to form a structured gating mechanism, coupled with a temperature-calibrated scoring head for stable optimization and reliable score distributions.

EdgeDAM Submitted May 2026

BMVC 2026 - British Machine Vision Conference - Results

- Proposed **EdgeDAM: Distractor-Aware Real-time Object Tracking for Mobile Devices.**
- EdgeDAM, a lightweight and mobile-friendly framework for single-object tracking. EdgeDAM integrates a fast YOLOv11s detection backbone, the Channel and Spatial Reliability Tracker (CSRT), built on discriminative correlation filters, and an enhanced Distractor-Aware Memory (DAM) module specifically optimized for detection-driven occlusion handling.

Lightweight Deep Learning for Visual Perception: A Survey Accepted August 2025

Neurocomputing

- Proposed **Lightweight Deep Learning for Visual Perception: A Survey of Models, Compression Strategies, and Edge Deployment Challenges.**
- This proposed survey systematically explores recent advancements in LDL models, highlighting their architectures, optimization techniques, and real-world applications.

DSConvNet Accepted December 2025

IEEE Access

- Proposed **DSConvNet: A Lightweight Depthwise Separable Convolution Network for Multiclass Image Feature Extraction.**
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UxV-Swarm using Instance Segmentation Accepted April 2024

The 15th International Conference on Ubiquitous and Future Networks (ICUFN)

- Proposed **UxV-Swarm based Automated Fruits Disease Instance Segmentation for Smart Farming using Deep Learning Model.**

Real-Time CCTV Based Garbage Detection Accepted March 2023

Journal of Information and Communication Convergence Engineering (JICCE)

- Proposed **Real-Time CCTV Based Garbage Detection for Modern Societies using Deep Convolutional Neural Network with Person-Identification**

2nd Author Publications

Uncertainty-Aware and Decoder-Aligned Learning for Video Summarization Accepted March 2026

IJCNN - International Joint Conference on Neural Networks

- Proposed VASTSum, an uncertainty-aware and decoder-aligned learning framework for video summarization that addresses both challenges within a single-pass model. The proposed method predicts probabilistic frame-level importance scores using a variational formulation, enabling explicit modeling of uncertainty arising from multi-annotator supervision. To account for subjectivity, particularly under binary annotations, we employ a supervision strategy that encourages alignment with plausible human annotation modes rather than enforcing a single consensus target.
- Furthermore, we introduce a decoder- aligned regularization that promotes stability of knapsack-based summary selection, reducing sensitivity to small perturbations in predicted scores.

SafeVision Accepted December 2025

Neurocomputing

- Proposed **This paper presents SafeVision a unified vision–language framework for real-time, context-aware safety assessment.**
- The SafeVision integrates a frozen CLIP-ViT encoder with two complementary reasoning modules: the Scene-Aware Reasoner (SAR) for global environmental context understanding and Region-Focused Neural Reasoner for region-level safety compliance verification.

CrossFusionKD Submitted November 2025

IEEE IoT Journal

- Proposed **CrossFusion Knowledge Distillation (CrossFusionKD) framework enhances detection performance while reducing computational overhead by leveraging model compression and cross-modal distillation.**
- A teacher model trained on RGB and IR data transfers rich multi-modal knowledge to a lightweight, RGB-only student model through channel-wise distillation, enabling efficient multi-modal knowledge transfer without requiring IR sensors during inference.

Projects

Large-Scale 2D Auto-Labeling and Keyframe Selection for Robotics Datasets

- Designed an end-to-end 2D auto-labeling pipeline for large-scale robotic perception datasets, enabling efficient pseudo-label generation for object detection models.
- Developed a keyframe selection strategy using DINOv3 self-supervised visual representations to identify semantically diverse and informative frames, reducing redundancy in long video sequences.
- Leveraged DINOv3 feature similarity and temporal consistency to filter near-duplicate frames, improving dataset diversity and annotation efficiency for industrial-scale data collection.
- Integrated the selected keyframes with a 2D object detection auto-labeler to generate high-quality pseudo-labels, significantly minimizing manual annotation cost while preserving training performance.
- Applied the pipeline to real-world robot-collected data, supporting scalable dataset understanding and continuous model improvement in autonomous navigation scenarios.

3D Pose Estimation and Avatar Generation

- Developed a 3D human pose estimation pipeline for avatar generation using monocular RGB input.
- Employed 2D pose and object detection models to extract joint keypoints and structural cues from video frames.
- Lifted 2D predictions into 3D space via RGB-based depth reasoning and point projection to generate coarse 3D meshes.
- Applied human motion and texture modeling techniques to generate temporally consistent 3D avatars suitable for visualization and animation tasks.

Motion Categorization using Generative AI Models

- Developed a real-time 2D motion categorization framework for sports applications, focusing on dynamic trajectory understanding and prediction.
- Generated 2D object tracks and depth maps using off-the-shelf detection and depth estimation models as input to a motion encoder.
- Designed a track encoder–decoder architecture that integrates DINO visual features to decouple motion dynamics from semantic appearance.
- Utilized SAM2 to group dynamic trajectories belonging to the same object and generate fine-grained moving object masks for improved motion segmentation.

3D Mesh Simplification and Compression Analysis

- Developed an automated pipeline for 3D mesh simplification and compression analysis across multiple decimation levels.
- Implemented a Python–Open3D framework to batch-process polygonal meshes (OBJ/PLY/STL/OFF) using quadric-error-based decimation at varying target triangle ratios.
- Designed evaluation metrics to quantify compression–quality trade-offs, including vertex and triangle reduction, file size compression ratio, and approximate Chamfer distance.
- Built an automated reporting system exporting per-model JSON summaries, enabling efficient comparison of geometric distortion across simplification levels.
- Optimized mesh preprocessing with degenerate triangle removal and normal recomputation, improving robustness for large-scale mesh datasets.

Certificates

Streamlined Data Ingestion with pandas *Jan 2023*
DataCamp

Neural Networks and Deep Learning *Mar 2023*
Coursera

Writing Efficient Python Code *May 2023*
DataCamp

Languages

English *Level: Advanced*

Korean *Level: Basic*

References

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